# BAHIRDAR INSTITUTE OF TECHNOLOGY

# FUCULTY OF COMPUTING

# DEPARTMENT OF SOFTWARE ENGINEERING

## *NETRUNNER OPRATING SYSTEM*

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**COURSE TITLE: OPRATING SYSTEM**

**SECTION: A**

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**SUBMISSION DATE: APRIL 28/04/2026**

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## 1. Introduction (Background & Motivation)

### i. Overview of Operating Systems

An Operating System (OS) is the most essential system software that manages computer hardware and provides a platform for application programs to run. It acts as an interface between the user and the computer hardware, ensuring efficient utilization of resources such as the processor, memory, storage, and input/output devices. Without an operating system, it would be impossible for users to interact with a computer system in a meaningful way.

Modern operating systems such as Microsoft Windows, Linux distributions, and macOS are designed to support multitasking, multi-user environments, process scheduling, memory management, file system organization, networking, and security. Over time, operating systems have evolved significantly to meet the growing demands of users and advancements in hardware technologies.

Operating systems can be classified into several categories based on their functionality and usage:

- **Single-user and Multi-user Operating Systems:**
  These determine whether one or multiple users can access the system simultaneously.
- **Real-Time Operating Systems (RTOS):**
  Used in time-critical applications such as robotics, aviation systems, and industrial automation.
- **Distributed Operating Systems:**
  Manage a group of independent computers and make them appear as a single system.
- **Embedded Operating Systems:**
  Designed for specialized devices such as smartphones, IoT devices, and embedded systems.
- **General-Purpose Operating Systems:**
  Used in personal computers and servers, providing flexibility for a wide range of applications.

In this project, the selected operating system is **NetRunner**, a Linux-based distribution built on Ubuntu. NetRunner is known for its user-friendly interface, stability, and compatibility with a wide range of software packages. Because it is based on Ubuntu

(which has Long-Term Support – LTS), it provides reliability, security updates, and strong community support.

NetRunner is particularly suitable for educational purposes because it combines the power of Linux with an easy-to-use desktop environment, making it ideal for learning system programming, file management, and OS configuration.

**Virtualization Concept**

Virtualization is one of the most important technologies in modern computing. It allows a single physical machine to run multiple operating systems simultaneously by creating virtual environments that simulate real hardware. Each virtual environment, known as a **virtual machine (VM)**, operates independently with its own OS, applications, and resources.

Virtualization is implemented using software called a **hypervisor**, which manages and allocates hardware resources to each virtual machine. Hypervisors are classified into two main types:

- **Type 1 (Bare-metal Hypervisors):**
  These run directly on the physical hardware without a host operating system. Examples include VMware ESXi and Microsoft Hyper-V.
- **Type 2 (Hosted Hypervisors):**
  These run on top of an existing operating system and are easier to use for learning and experimentation. Examples include VMware Workstation and Oracle VirtualBox.

In this project, **VMware Workstation** is used as the virtualization platform. It is a Type 2 hypervisor that allows users to create and manage virtual machines efficiently. It provides features such as snapshot management, virtual networking, hardware customization, and easy OS installation, making it highly suitable for academic and practical learning environments.

**Benefits of Virtualization**

Virtualization provides several important advantages in modern computing:

- **Safe Environment:**
  Users can install and test operating systems without affecting the host system.
- **Cost Efficiency:**
  Multiple virtual machines can run on a single physical machine, reducing hardware costs.

- **Flexibility:**
  Virtual machines can be easily created, modified, copied, or deleted.
- **System Recovery:**
  Snapshots allow users to restore the system to a previous state in case of failure.
- **Isolation:**
  Problems in one virtual machine do not affect others or the host system.

**ii. Motivation for the Project**

The motivation for this project is based on several academic and practical factors:

1. **Practical Learning Experience:**
   Installing NetRunner in a virtual environment provides hands-on experience in operating system installation, configuration, and management. It helps bridge the gap between theoretical knowledge and real-world application.
2. **Understanding Linux-Based Systems:**
   NetRunner, being a Linux distribution, allows exploration of Linux file systems, command-line tools, and system administration concepts, which are essential skills in modern computing.
3. **Development of Virtualization Skills:**
   Working with VMware Workstation helps in understanding how virtual machines are created, configured, and managed. These skills are widely used in IT industries, cloud computing, and software development.
4. **Problem-Solving and Troubleshooting:**
   During the installation process, challenges such as screenshot issues and configuration problems may arise. Solving these problems improves analytical thinking and technical skills.
5. **Safe Experimentation:**
   Virtualization allows experimenting with operating systems without risking damage to the host system. This encourages learning through trial and error.
6. **Preparation for Future Careers:**
   Knowledge of operating systems and virtualization is essential for careers in software engineering, system administration, cybersecurity, and cloud computing.

## 2. Objectives

The objectives of this project are designed to provide a comprehensive understanding of operating systems, virtualization, and system programming through both theoretical knowledge and practical implementation. By installing and working with NetRunner in a virtual environment using VMware Workstation, the project aims to achieve the following goals:

### 1. Practical Operating System Installation Skills

- To learn how to install a Linux-based operating system (**NetRunner**) in a virtualized environment using VMware Workstation.
- To understand the complete installation process, including selecting the ISO file, configuring virtual hardware, disk partitioning, and system setup.
- To gain experience in creating and managing virtual machines, including allocating CPU, RAM, and storage resources.
- To follow proper installation procedures and ensure correct user account creation using the full name as required.

### 2. Understanding Linux-Based Operating Systems

- To explore the structure and functionality of a Linux-based OS, including file management, user interface, and system utilities.
- To understand how Linux distributions like NetRunner are built on top of Ubuntu and benefit from long-term support (LTS).
- To study core OS concepts such as process management, memory management, and system services within a Linux environment.
- To gain familiarity with both graphical and command-line interactions in Linux systems.

### 3. Understanding Virtualization Concepts

- To analyze how virtualization works and how VMware Workstation creates a virtual hardware environment.
- To understand the role of a hypervisor and the difference between Type 1 and Type 2 virtualization.
- To learn how system resources are shared between the host operating system and the virtual machine.

- To explore features such as snapshots, virtual networking, and hardware abstraction in virtualization.

**4. Filesystem Knowledge and Analysis**

- To identify the filesystem used by NetRunner (such as ext4) and understand its structure and functionality.
- To study important filesystem features such as journaling, data integrity, and performance optimization.
- To compare Linux filesystems (ext4, Btrfs, ZFS) with other filesystems like NTFS, FAT32, exFAT, HFS+, and APFS.
- To understand how filesystem selection affects system performance, reliability, and compatibility.

**5. Troubleshooting and Problem-Solving Skills**

- To identify and document issues encountered during installation, such as screenshot limitations or configuration problems.
- To develop practical solutions for resolving installation and system-related errors.
- To improve analytical thinking and debugging skills through real-world problem-solving.
- To learn how to document technical issues and their solutions clearly and professionally.

**6. Academic and Documentation Skills**

- To develop structured technical documentation skills by preparing a detailed report of the entire project.
- To present installation steps, issues, solutions, and analysis in a clear and organized manner.
- To integrate theoretical concepts of operating systems with practical implementation.
- To improve technical writing skills suitable for academic and professional purposes.

**7. Preparation for Real-World Applications**

- To gain experience with virtualization tools commonly used in industry environments.

- To understand how virtual machines are used in software development, testing, and system administration.
- To build foundational knowledge useful for careers in IT, software engineering, cybersecurity, and cloud computing.
- To prepare for advanced topics such as server virtualization, cloud platforms, and distributed systems.

**3. Requirements**

To successfully install and run NetRunner in a virtual environment using VMware Workstation, both hardware and software requirements must be satisfied. These requirements ensure smooth performance of the virtual machine while maintaining stability of the host system. Since virtualization involves sharing physical resources between the host and guest operating systems, proper configuration is essential for efficient operation.

**3.1 Hardware Requirements**

Although NetRunner is a lightweight Linux-based operating system, running it inside a virtual machine requires sufficient hardware resources from the host system. The following specifications were used and are recommended:

**1. Processor (CPU):**

- **Specification Used:** Intel Core i7 (64-bit processor)
- **Minimum Requirement:** Intel Core i3 or equivalent
- **Recommended:** Intel Core i5/i7 or AMD Ryzen series

**Explanation:**
The processor is responsible for executing instructions for both the host operating system and the virtual machine. A multi-core processor such as Intel Core i7 ensures that virtualization tasks are handled efficiently without slowing down the system. Since VMware Workstation distributes workloads between the host and guest OS, a powerful CPU significantly improves performance.

**2. Memory (RAM):**

- **Specification Used:** 16 GB system memory
- **Minimum Requirement:** 4 GB
- **Recommended Allocation for VM:** 4–8 GB

**Explanation:**
RAM is one of the most critical resources in virtualization. The host system must have enough memory to run both Windows 10 and the NetRunner virtual machine simultaneously. Allocating sufficient RAM to the VM ensures smooth installation and operation without lag or crashes.

**3. Storage (Hard Disk):**

- **Specification Used:**
  - Total Storage: 512 GB
  - Virtual Disk Allocated: 20 GB
- **Minimum Requirement:** 20 GB free disk space
- **Recommended:** 30 GB or more for additional applications

**Explanation:**
Storage is required to install the operating system and store files within the virtual machine. A 20 GB virtual disk is sufficient for NetRunner installation and basic usage. Using a single virtual disk file improves performance and simplifies management.

**4. Host Machine:**

- **Specification Used:** Laptop running Windows 10

**Explanation:**
The host operating system provides the platform on which VMware Workstation runs. Windows 10 supports virtualization technologies and is compatible with VMware, making it suitable for running virtual machines efficiently.

**5. Network Adapter:**

- **Specification Used:** Wireless network adapter

**Explanation:**
Network connectivity is necessary for downloading updates, installing software packages, and accessing online resources within the virtual machine. VMware provides virtual networking options (NAT or Bridged) to enable internet access.

**6. Additional Hardware Considerations:**

- Keyboard and mouse are managed through VMware's virtual interface.
- Display resolution should support at least 1024×768 for proper GUI display.

- Virtual GPU provided by VMware ensures graphical interface functionality.

**3.2 Software Requirements**

**1. Virtualization Platform:**

- **Software Used:** VMware Workstation

**Explanation:**
VMware Workstation is a Type 2 hypervisor that allows users to create and run virtual machines on top of a host operating system. It provides advanced features such as:

- Virtual hardware configuration
- Snapshot and recovery options
- Virtual networking
- Easy OS installation

This makes it ideal for educational and experimental purposes.

**2. Guest Operating System:**

- **Operating System Used:** NetRunner (Linux-based, Ubuntu LTS)

**Explanation:**
NetRunner is a user-friendly Linux distribution that offers stability, security, and compatibility with a wide range of applications. Being based on Ubuntu LTS ensures long-term support and regular updates, making it reliable for learning and development.

**3. ISO Image File:**

- NetRunner ISO file is required for installation.

**Explanation:**
The ISO file contains the complete installation package of the operating system. It is loaded into VMware Workstation to boot and install the OS inside the virtual machine.

**4. VMware Tools (Optional):**

- Used to improve VM performance and usability.

**Features Include:**

- Better screen resolution
- Improved graphics performance

- Shared clipboard between host and VM
- Drag-and-drop functionality

**5. Documentation and Screenshot Tools:**

- Screenshot tools (built-in or external) for capturing installation steps
- In this project, some screenshots were captured using a mobile phone due to limitations during installation

**3.3 Notes on System Setup**

- Ensure that virtualization is enabled in BIOS/UEFI settings.
- Allocate resources carefully to avoid overloading the host system.
- Maintain sufficient free disk space for smooth VM performance.
- Use updated versions of VMware Workstation for compatibility and stability.

**Summary**

Meeting these hardware and software requirements ensures:

- Smooth installation and operation of NetRunner
- Efficient resource allocation between host and virtual machine
- Ability to explore OS features, filesystem behavior, and virtualization concepts without performance

**4. Installation Steps of NetRunner in VMware Workstation**

**4.1 Launch VMware Workstation**

1. Open VMware Workstation on the host computer (Windows 10).
2. From the main interface, click on **"Create a New Virtual Machine."**
3. In the setup wizard, select **"Typical (recommended)"** configuration.
4. Click **Next** to proceed.

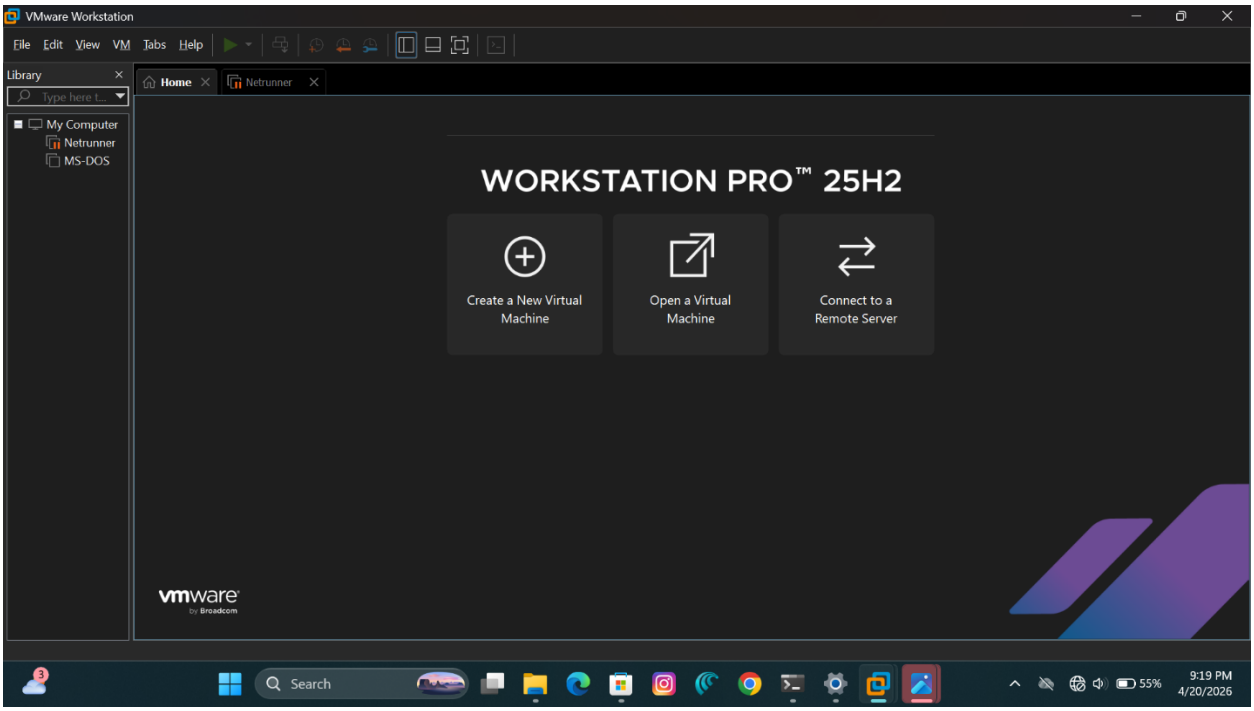



Figure 1: Starting VMware Workstation and selecting “Create a New Virtual Machine.”

### 4.2 Select the NetRunner ISO File

1. In the installation options, choose **“Installer disc image file (ISO)”**.
2. Click **Browse** and locate the NetRunner ISO file.
3. Select the ISO file and click **Next**.

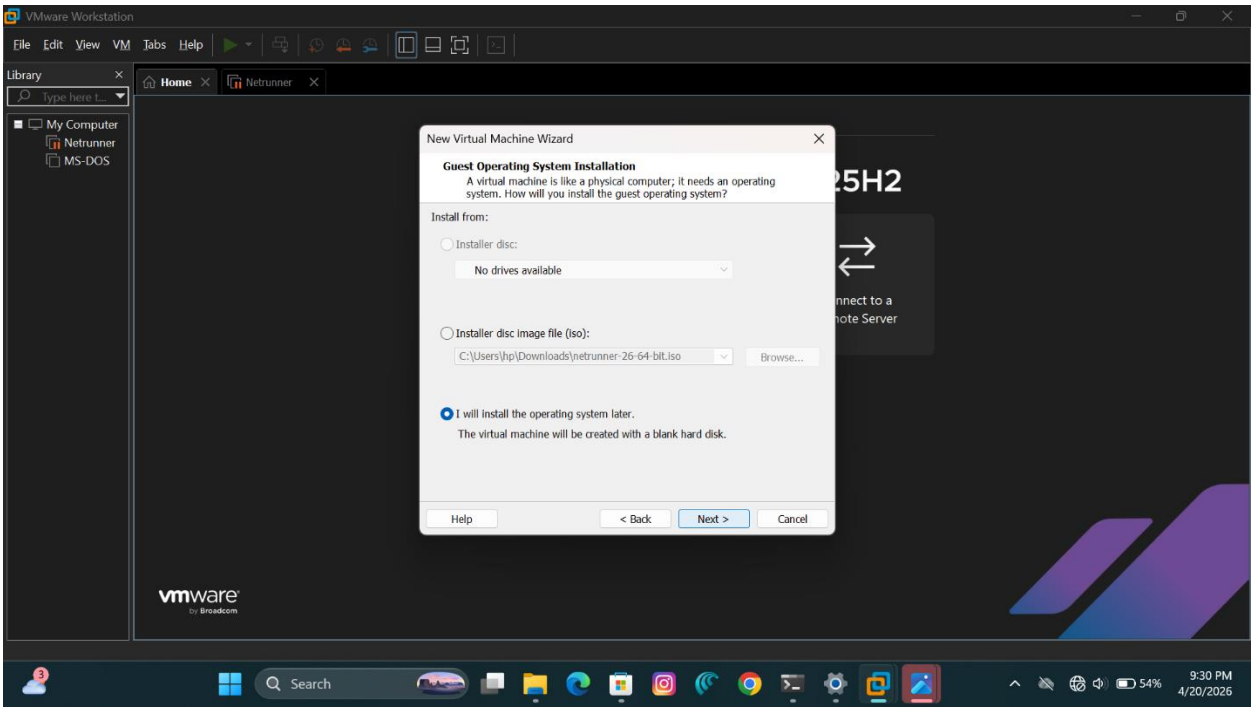



Figure 2: Selecting the NetRunner ISO file for installation.

### 4.3 Name the Virtual Machine and Choose Location

1. Enter the name of the virtual machine using your **full name**.
2. Choose a location on the host system where VM files will be stored.
3. Click **Next** to continue.

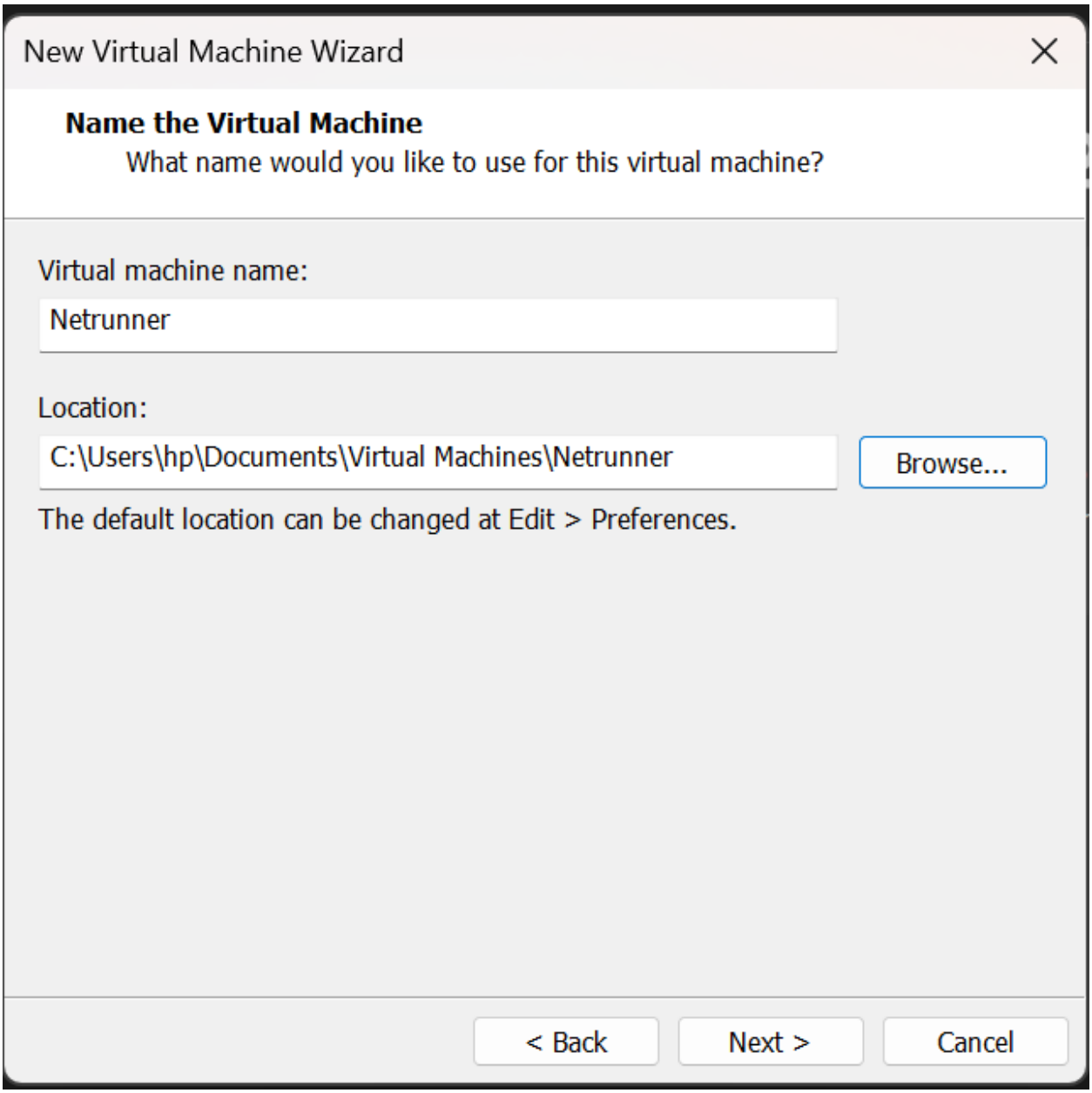



Figure 3: Naming the virtual machine and selecting storage location.

**4.4 Configure Virtual Hardware**

1. Set the maximum disk size to **20 GB**.
2. Choose **"Store virtual disk as a single file."**
3. Allocate system resources:
    - RAM: 4–8 GB
    - CPU: 1–2 cores

4. Ensure the network adapter is enabled (NAT recommended).
5. Click **Finish** to create the virtual machine.

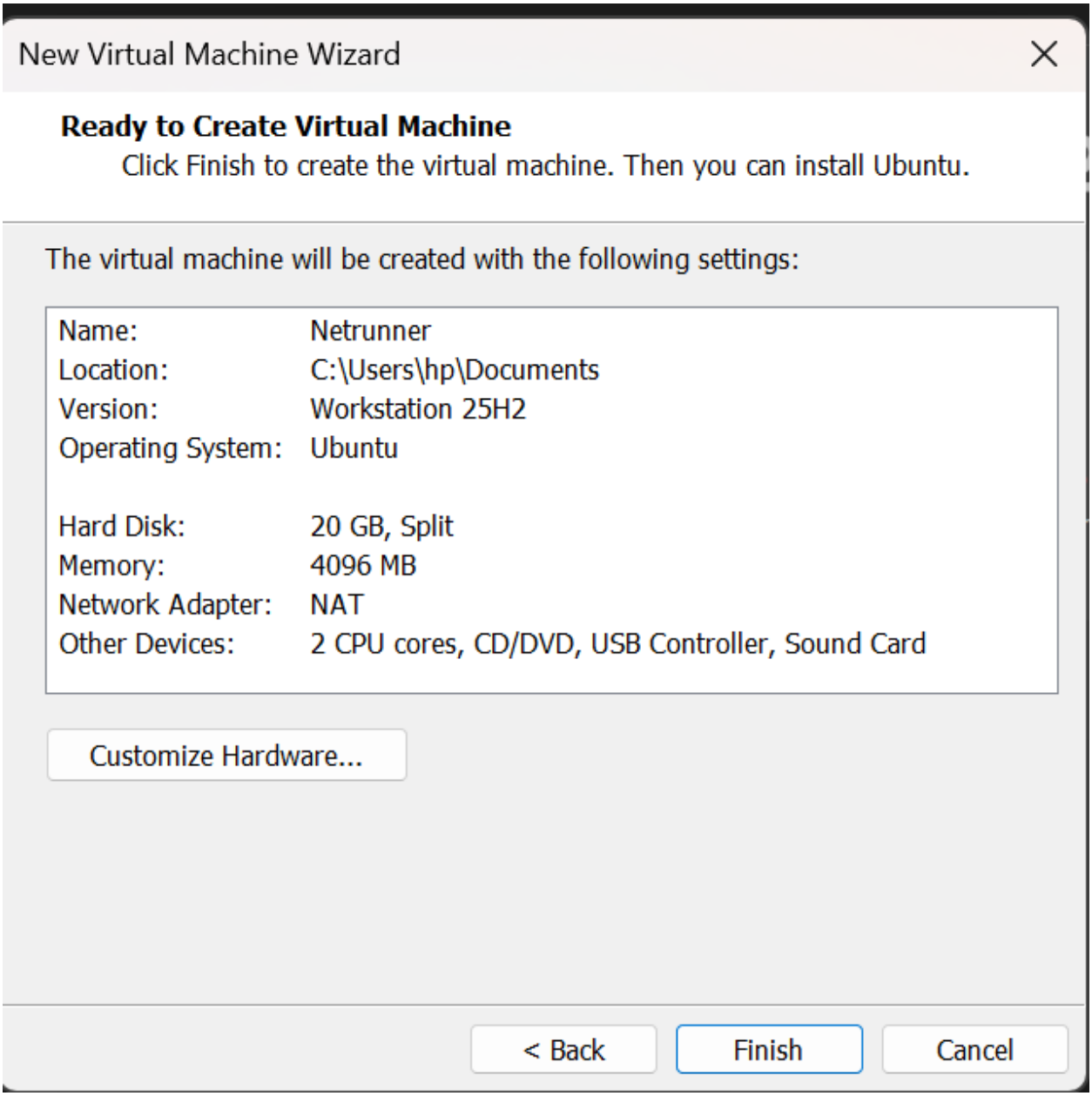



Figure 4: Configuring virtual hardware (RAM, CPU, disk size).

**4.5 Start the Virtual Machine and Boot the Installer**

1. Select the created virtual machine.
2. Click **"Power on this virtual machine."**
3. The system will boot from the NetRunner ISO file.

4. Select **“Start NetRunner Installation.”**

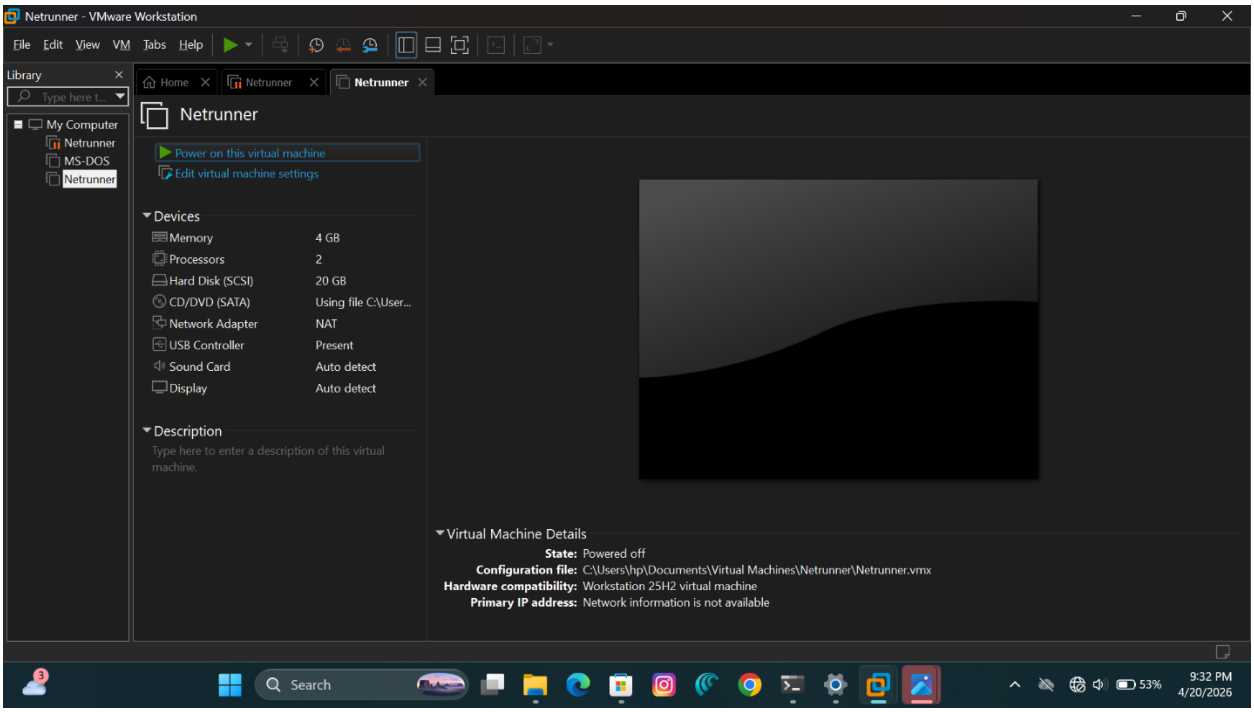
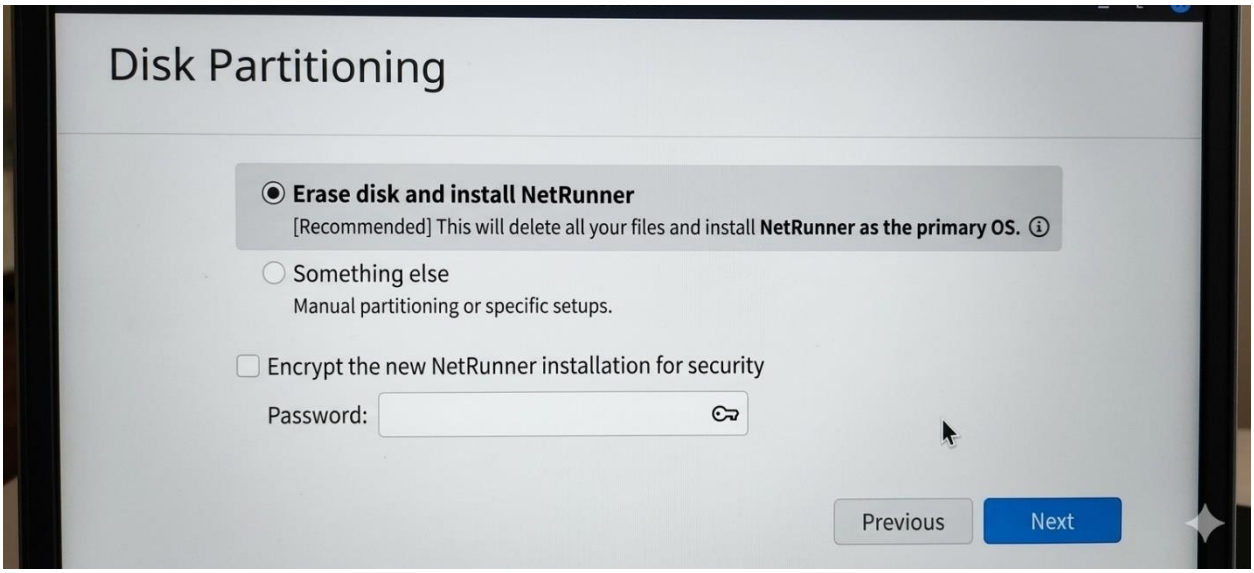



Figure 5: NetRunner boot menu during installation.

### 4.6 Disk Partitioning and Installation Setup

1. Choose the installation type:
    - Select **“Erase disk and install NetRunner”** for automatic setup.
2. Optionally enable encryption if required.
3. Proceed to the next step.




**Caption:** Figure 6: Disk partitioning and installation setup.

### 4.7 Create User Account (Using Full Name)

1. Enter your **full name** as required.
2. Set a username (if not automatically generated).
3. Enter and confirm a secure password.
4. Click **Continue** to proceed.

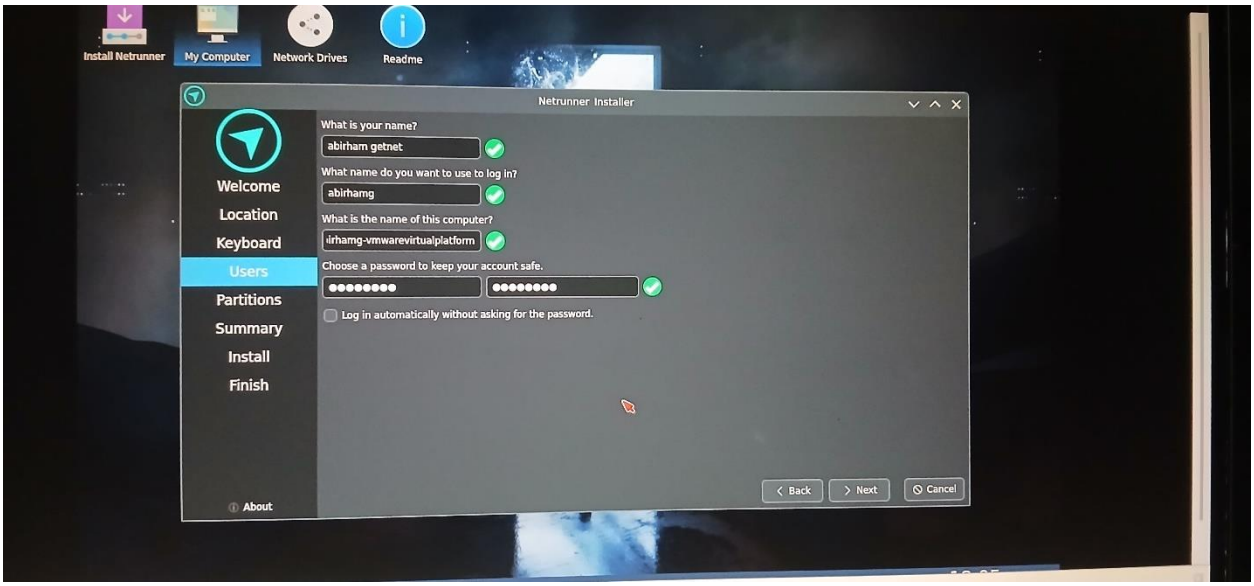



Figure 7: Creating user account using full name

### 4.8 Installation Progress

1. The system will begin copying files and installing NetRunner.
2. Wait for the installation process to complete.

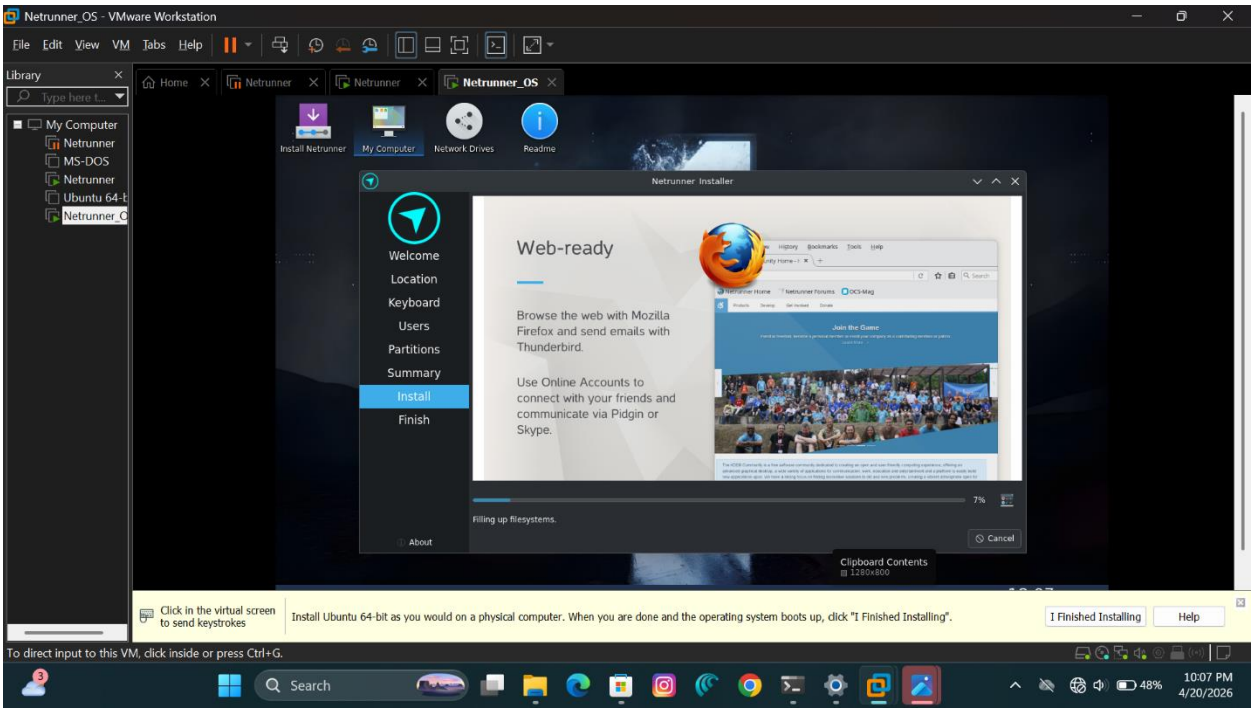



Figure 8: NetRunner installation in progress.

### 4.9 Complete Installation and Reboot

1. Once installation is complete, click **Restart Now**.
2. Remove the ISO file if prompted.
3. The system will reboot into NetRunner.
4. Log in using the created account.

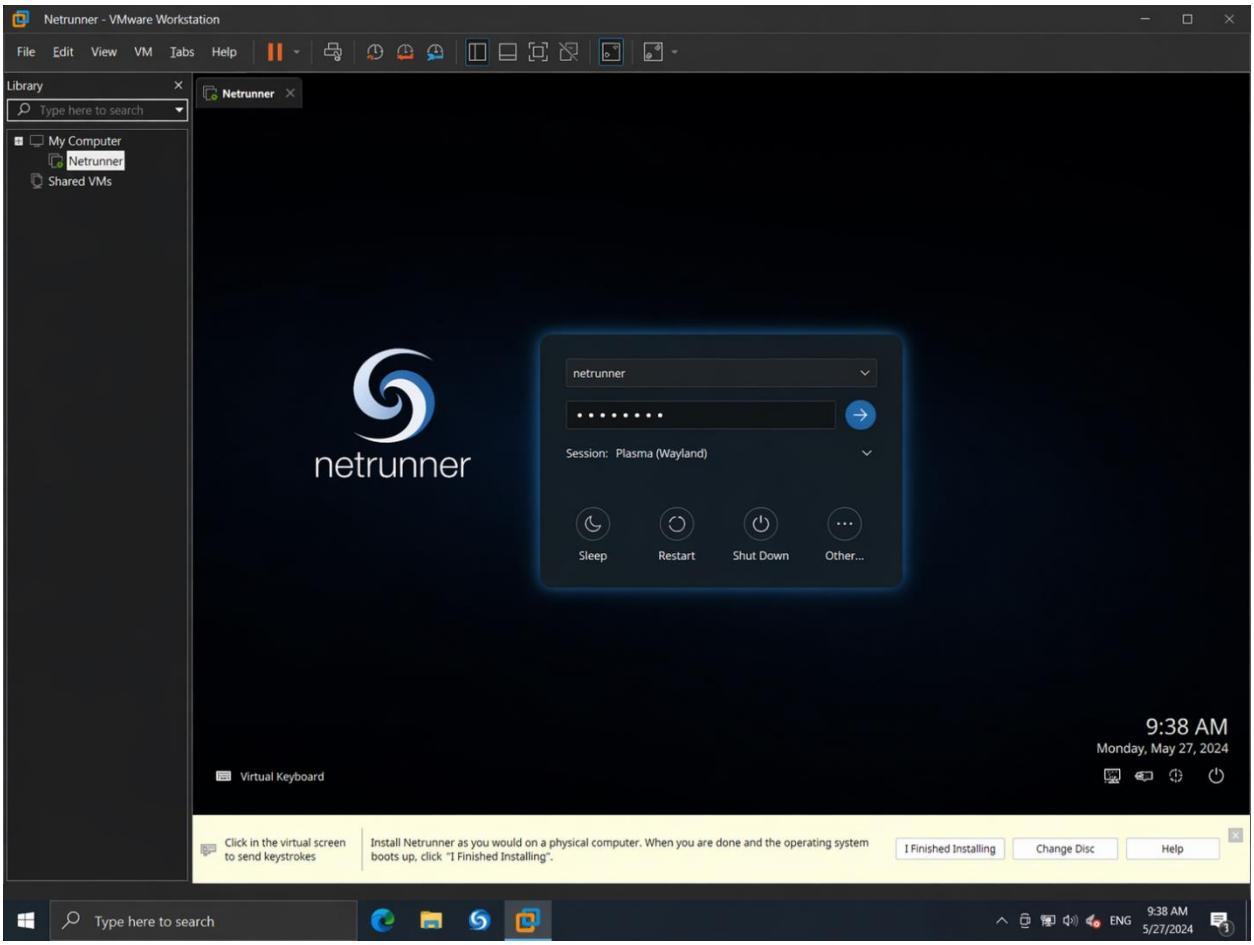



Figure 9: NetRunner login screen after installation.

### 4.10 First Boot and Desktop Environment

1. After logging in, the NetRunner desktop environment will load.
2. Verify that the system is functioning properly.

**Caption:** Figure 10: NetRunner desktop after successful installation.

## 5. Issues (Problems Faced)

During the installation of NetRunner in VMware Workstation, several practical challenges were encountered. These issues were primarily related to system limitations, virtualization environment behavior, and documentation difficulties. Identifying and documenting these problems is important for understanding real-world installation scenarios and improving troubleshooting skills.

### 5.1 Screenshot Capture Limitation During Installation

**Problem Description:**
While performing the installation, screenshots were required to document each step. However, during certain stages—especially at the **user account creation step (entering full name and password)**—the system did not allow screenshots to be captured properly.

**Observation:**

- The screenshot tool did not respond or failed to save images.
- This issue occurred specifically during sensitive input stages such as password entry.
- Screenshots taken earlier or later worked normally.

**Possible Cause:**

- The NetRunner installer runs in a temporary live environment with limited permissions.
- Security restrictions may prevent screenshot capture during password input to protect user data.
- VMware display integration may not fully support screenshot tools during certain installation phases.

**Impact:**

- Incomplete documentation of some installation steps.
- Required alternative methods to capture important stages.

### 5.2 Screenshot Storage Location Confusion

**Problem Description:**
Screenshots taken inside the NetRunner virtual machine were not found in the expected location on the host system (Windows 10).

**Observation:**

- Screenshots were saved inside the NetRunner OS instead of the host OS.
- The user could not easily locate or transfer them to Windows.

**Possible Cause:**

- Virtual machines operate independently, so files created inside the VM are stored in the VM's filesystem.
- Lack of shared folders or integration tools (such as VMware Tools) prevented easy file transfer.

**Impact:**

- Difficulty accessing screenshots for report preparation.

- Increased time spent locating and transferring files.

### 5.3 Limitation of Screenshot Tools in Virtual Environment

**Problem Description:**
Some screenshot tools did not function correctly during the installation process inside the virtual machine.

**Observation:**

- Screenshot tools worked inconsistently.
- Some steps could not be captured at all.

**Possible Cause:**

- Limited functionality of screenshot utilities in live installer environments.
- Lack of full OS functionality before installation is completed.

**Impact:**

- Forced reliance on external methods (e.g., phone camera).
- Reduced quality and consistency of captured images.

## 6. Solutions

For each problem encountered during the installation process, appropriate solutions or workarounds were applied to ensure successful completion of the project.

### 6.1 Using External Device for Screenshot Capture

**Solution:**

- A mobile phone camera was used to capture images of the screen during steps where screenshots could not be taken.

**Explanation:**

- This method ensured that all required installation steps were documented, even when internal tools failed.
- Although not ideal, it is a practical workaround in restricted environments.

### 6.2 Understanding File Storage in Virtual Machines

**Solution:**

- Recognized that screenshots taken inside NetRunner are stored within the virtual machine's filesystem.
- Accessed files directly within NetRunner or used alternative methods to retrieve them.

**Recommended Improvement:**

- Install VMware Tools to enable:
    - Shared folders
    - Drag-and-drop file transfer
    - Clipboard sharing

**6.3 Alternative Screenshot Methods**

**Solution:**

- Used host system screenshot methods (e.g., **Print Screen key**) where possible.
- Took screenshots at stages where the system allowed capture.

**Explanation:**

- Combining multiple methods helped ensure that most steps were documented.

**6.4 General Troubleshooting Approach**

- Carefully observed system behavior during installation.
- Identified patterns where issues occurred.
- Applied logical solutions based on understanding of virtualization and OS behavior.

**Summary of Issues and Solutions**

The problems encountered were mainly related to **virtualization limitations and system restrictions during installation**. Despite these challenges, practical solutions were applied successfully.

This experience highlights the importance of:

- Flexibility in problem-solving
- Understanding system behavior in virtual environments
- Using alternative approaches when standard tools fail

## 7. Filesystem Support

Understanding filesystems is a fundamental part of operating systems and system programming. A filesystem defines how data is stored, organized, accessed, and managed on storage devices. Different operating systems support different filesystems, each optimized for specific performance, reliability, and compatibility requirements.

NetRunner, being a Linux-based operating system, supports multiple modern filesystems, with **ext4** as the default and most commonly used option. This section explains the selected filesystem, its features, and compares it with other widely used filesystems.

### 7.1 Primary Filesystem Used: ext4 (Fourth Extended Filesystem)

**Overview:**

The **ext4 (Fourth Extended Filesystem)** is the default filesystem used by most modern Linux distributions, including NetRunner. It is an improvement over earlier versions (ext2 and ext3) and is designed to provide high performance, reliability, and scalability.

**Key Features of ext4:**

✓ **Journaling Support:**

- Ext4 uses a journaling mechanism to record changes before they are applied.
- This helps prevent data corruption in case of system crashes or power failures.

✓ **Large File and Volume Support:**

- Supports very large file sizes (up to 16 TB) and filesystem sizes (up to 1 EB).
- Suitable for both small systems and large-scale storage environments.

✓ **Improved Performance:**

- Uses techniques such as delayed allocation and multiblock allocation.
- Reduces fragmentation and improves read/write speed.

✓ **Backward Compatibility:**

- Compatible with ext2 and ext3, allowing easy upgrades.

✓ **Reliability and Stability:**

- Widely tested and used, making it one of the most stable Linux filesystems.

**Reason for Selecting ext4:**

- Default and recommended filesystem for NetRunner
- High stability and reliability
- Good performance in virtual environments
- Wide support across Linux systems

**7.2 Other Supported Filesystems in NetRunner**

NetRunner also supports several other filesystems, each with unique features:

**1. Btrfs (B-Tree Filesystem)**

- Advanced Linux filesystem with modern features
- Supports **snapshots**, **compression**, and **self-healing**
- Suitable for advanced users and system administrators

**Limitation:** Slightly complex and not as stable as ext4 in all cases

**2. ZFS (Zettabyte File System)**

- High-performance filesystem with data integrity features
- Supports **data redundancy**, **error correction**, and **snapshots**
- Used in enterprise and large storage systems

**Limitation:** Requires more memory and configuration

**3. NTFS (New Technology File System)**

- Default filesystem for Windows operating systems
- Supported in Linux for compatibility purposes

**Use Case:**

- Accessing Windows partitions from NetRunner

**4. FAT32 (File Allocation Table)**

- Older filesystem with high compatibility
- Used for USB drives and removable media

**Limitation:**

- Maximum file size of 4 GB

**5. exFAT (Extended File Allocation Table)**

- Designed for flash storage devices
- Supports larger files compared to FAT32

**6. HFS+ and APFS**

- Filesystems used in macOS systems
- Limited support in Linux through additional tools

**7.4 Observations**

- **ext4** is the best choice for NetRunner due to its balance of performance, stability, and simplicity.
- Advanced filesystems like **Btrfs** and **ZFS** provide additional features but are more complex.
- Filesystems like **NTFS, FAT32, and exFAT** are mainly used for compatibility with other operating systems.
- Modern filesystems focus on **data integrity, scalability, and performance optimization**.

**7.5 Summary**

The filesystem plays a critical role in operating system performance and reliability. In this project:

- **ext4** was selected as the primary filesystem due to its proven reliability and efficiency.
- Alternative filesystems provide specialized features for different use cases.
- Understanding filesystem differences helps in choosing the right system for specific needs.

**8. Advantages and Disadvantages of Installing NetRunner in VMware Workstation**

In this project, NetRunner was installed and tested within a virtual environment using VMware Workstation. This approach provided practical insights into both the benefits and

limitations of virtualization and Linux-based operating systems. The following points summarize the advantages and disadvantages observed during the installation and usage process.

### 8.1 Advantages

**1. Safe and Isolated Learning Environment**

- VMware Workstation provides a completely isolated environment where NetRunner can be installed without affecting the host operating system (Windows 10).
- Any errors, misconfigurations, or system failures inside the virtual machine do not impact the main system.
- This makes it ideal for students and beginners to experiment freely.

**2. Hands-On Experience with Linux Systems**

- Installing NetRunner provided practical exposure to Linux-based operating systems.
- It helped in understanding system installation, configuration, and user management.
- The experience also introduced Linux filesystem concepts and system behavior.

**3. Efficient Use of Hardware Resources**

- Virtualization allows multiple operating systems to run on a single physical machine.
- This reduces the need for additional hardware.
- Resources such as CPU, RAM, and storage can be allocated dynamically based on requirements.

**4. Flexibility and Easy Configuration**

- VMware Workstation allows easy modification of virtual machine settings such as memory, CPU cores, and disk size.
- Virtual machines can be created, deleted, or cloned quickly.
- This flexibility is useful for testing different configurations and setups.

**5. Snapshot and Recovery Features**

- VMware provides snapshot functionality, allowing users to save the state of a virtual machine at a specific point in time.

- If an issue occurs, the system can be restored to a previous working state بسهولة.
- This feature is very useful for experimentation and learning.

**6. Compatibility and Portability**

- Virtual machines can be easily transferred between systems.
- NetRunner can run on different host operating systems without compatibility issues.
- This makes virtualization a powerful tool for development and testing environments.

**7. Development of Troubleshooting Skills**

- Encountering issues such as screenshot limitations and file access problems helped improve problem-solving abilities.
- The process of identifying and resolving issues enhanced practical technical skills.

**8.2 Disadvantages**

**1. Performance Overhead**

- Running NetRunner inside VMware Workstation consumes system resources such as CPU and RAM.
- The performance of the virtual machine is slightly lower compared to installing the OS directly on physical hardware.
- Heavy applications may run slower in a virtual environment.

**2. Limited Access to Hardware Features**

- Some hardware components (such as advanced graphics processing or certain peripherals) may not be fully accessible inside the virtual machine.
- The VM uses virtualized hardware instead of real physical components.

**3. Storage Consumption**

- Virtual machines require dedicated disk space for storing OS files and data.
- The allocated 20 GB virtual disk occupies space on the host system.
- Managing multiple virtual machines can quickly consume large storage capacity.

**4. Screenshot and Integration Limitations**

- As experienced in this project, some installation steps did not allow screenshots to be captured properly.
- Files created inside the VM are not directly accessible from the host without additional configuration (e.g., shared folders).

**5. Dependence on Host System**

- The performance and stability of the virtual machine depend on the host system.
- If the host system has limited resources or is overloaded, the VM performance will degrade.

**6. Learning Curve**

- Understanding virtualization concepts, configuring VMware, and managing Linux systems may be challenging for beginners.
- Additional time is required to learn both the virtualization platform and the operating system.

**8.3 Observations**

- Virtualization provides a powerful and flexible environment for learning and experimentation.
- NetRunner performed efficiently within the virtual machine when sufficient resources were allocated.
- Despite some limitations, the overall experience was highly beneficial for understanding operating systems and virtualization concepts.

**Summary**

The use of VMware Workstation to install NetRunner offers significant advantages in terms of safety, flexibility, and learning opportunities. However, it also introduces some limitations related to performance, hardware access, and system integration.

Overall, the advantages outweigh the disadvantages, making virtualization an effective approach for academic and practical exploration of operating systems.

## 9. Conclusion

In this project, the installation and exploration of the NetRunner operating system using VMware Workstation were successfully completed. The project provided a comprehensive understanding of operating systems, virtualization, and system programming concepts through both theoretical study and practical implementation.

### 9.1 Summary of the Work

The project began with understanding the fundamentals of operating systems and virtualization. A virtual machine was created using VMware Workstation, and NetRunner, a Linux-based operating system, was installed using an ISO image.

The installation process involved:

- Creating and configuring a virtual machine
- Allocating system resources such as CPU, RAM, and storage
- Performing disk partitioning and system setup
- Creating a user account using the full name as required
- Completing installation and verifying system functionality

All steps were carefully documented with explanations and screenshots (where possible), ensuring a clear and structured report.

### 9.2 Key Learning Outcomes

**1. Practical Operating System Installation Skills**

- Gained hands-on experience in installing a Linux-based operating system in a virtual environment.
- Learned how to configure and manage virtual machines effectively.
- Understood the importance of proper resource allocation for system performance.

**2. Understanding of Linux Systems**

- Explored the features and functionality of NetRunner as a Linux distribution.
- Learned about system configuration, user management, and desktop environment.
- Developed familiarity with Linux-based system operations.

**3. Virtualization Knowledge and Skills**

- Understood how virtualization works and how VMware Workstation functions as a Type 2 hypervisor.
- Learned how virtual hardware is created and managed.
- Gained experience in using virtualization tools for safe experimentation and testing.

**4. Filesystem Understanding**

- Studied the ext4 filesystem and its features such as journaling, performance optimization, and reliability.
- Compared ext4 with other filesystems like NTFS, FAT32, exFAT, Btrfs, and ZFS.
- Understood the role of filesystems in OS performance and data management.

**5. Problem-Solving and Troubleshooting**

- Encountered real-world issues such as screenshot limitations and file accessibility problems.
- Applied practical solutions and workarounds to overcome these challenges.
- Improved analytical thinking and troubleshooting skills.

**6. Documentation and Technical Writing Skills**

- Developed the ability to create structured and professional technical documentation.
- Learned how to present technical processes clearly with proper explanations and formatting.
- Improved academic writing skills suitable for system programming and IT-related projects.

### 9.3 Final Observations

- NetRunner proved to be a stable, user-friendly, and efficient Linux distribution suitable for learning and experimentation.
- VMware Workstation provided a reliable platform for virtualization, allowing safe and flexible OS installation.

- Virtualization plays a critical role in modern computing by enabling multiple operating systems to run on a single machine.
- Despite minor limitations, such as screenshot restrictions, the overall installation process was successful and educational.

### 9.4 Overall Conclusion

This project successfully demonstrated how operating systems can be installed and managed in a virtual environment. It highlighted the importance of virtualization in modern computing and provided practical exposure to Linux systems, filesystem concepts, and system configuration.

The combination of theoretical knowledge and hands-on practice made this project highly valuable in developing technical skills relevant to system programming, IT infrastructure, and software development.

## 10. Future Outlook / Recommendation

This project provided valuable experience in installing and working with a Linux-based operating system in a virtual environment. However, there are several areas where further exploration and improvement can be made. The following recommendations highlight potential future work and learning opportunities based on the outcomes of this project.

### 10.1 Enhancement of Virtualization Skills

- Future work can include deeper exploration of advanced features in VMware Workstation such as:
    - Snapshot management for multiple restore points
    - Cloning virtual machines
    - Advanced virtual networking (Bridged, NAT, Host-only)
- Understanding these features will improve the ability to manage complex virtual environments similar to real-world IT systems.

### 10.2 Exploration of Other Operating Systems

- Installing and comparing other operating systems can provide broader knowledge, such as:
    - Other Linux distributions (Ubuntu, Fedora, Kali Linux)

    - Windows-based systems
    - Legacy operating systems for historical understanding
- This will help in understanding differences in OS architecture, performance, and usability.

**10.3 Advanced Filesystem Analysis**

- Further study can be conducted on advanced filesystems such as:
    - **Btrfs** and **ZFS** for Linux
    - **NTFS** for Windows
    - **APFS** for macOS
- Performance comparison and benchmarking can be done to evaluate speed, reliability, and efficiency.

**10.4 Improvement in Documentation Techniques**

- Future projects can use better tools for documentation, such as:
    - Screen recording software
    - Advanced screenshot tools
    - Automated documentation tools
- This will help capture all steps clearly without missing important stages.

**10.5 Performance Optimization**

- Experimenting with different VM configurations can improve system performance, such as:
    - Increasing RAM allocation
    - Assigning more CPU cores
    - Adjusting virtual disk type and size
- This helps understand how resource allocation affects operating system performance.

**10.6 Learning System Administration and Networking**

- Future work can include:

    - Managing users and permissions in Linux
    - Configuring network settings in virtual machines
    - Installing and managing software packages
- These skills are essential for careers in system administration and IT infrastructure.

**10.7 Exploration of Cloud and Enterprise Virtualization**

- Virtualization can be extended to cloud computing platforms such as:
    - Virtual private servers (VPS)
    - Cloud environments (AWS, Azure, Google Cloud)
- Understanding cloud-based virtualization will prepare students for modern industry practices.

**10.8 Addressing Identified Limitations**

- The screenshot issue experienced during installation can be improved by:
    - Installing VMware Tools before capturing screenshots
    - Using host-based screenshot tools
    - Configuring shared folders for easy file transfer
- Addressing these limitations will improve future project quality.

**10.9 Overall Recommendation**

- Virtualization should be used as a standard approach for learning and experimenting with operating systems.
- Students are encouraged to practice installing multiple operating systems to strengthen their understanding.
- Combining theoretical knowledge with practical implementation is essential for mastering system programming concepts.

**Summary**

This project serves as a foundation for further learning in operating systems, virtualization, and system administration. By building on this experience, future work can expand into advanced topics such as cloud computing, distributed systems, and enterprise-level virtualization.

## 11. Virtualization in Modern Operating Systems

Virtualization is a fundamental technology in modern computing that allows efficient utilization of hardware resources by enabling multiple operating systems to run on a single physical machine. It plays a significant role in system programming, cloud computing, and IT infrastructure by improving flexibility, scalability, and performance.

This section explains virtualization in detail by addressing **what it is, why it is used, and how it works**.

### 11.1 What is Virtualization?

Virtualization is the process of creating a **virtual (rather than physical) version** of computing resources such as hardware platforms, operating systems, storage devices, and network resources. It allows a single physical system, known as the **host machine**, to run multiple independent environments called **virtual machines (VMs)**.

Each virtual machine behaves like a real computer system, with its own operating system, applications, and resources. Even though all VMs share the same physical hardware, they operate independently and are isolated from each other.

For example, in this project, NetRunner (Linux-based OS) was installed inside VMware Workstation on a Windows 10 system. Here:

- Windows 10 acts as the **host operating system**
- VMware Workstation acts as the **virtualization platform (hypervisor)**
- NetRunner acts as the **guest operating system**

This setup allows the user to run both operating systems simultaneously without interference.

### 11.2 Why Virtualization is Used

Virtualization is widely used in modern computing environments due to its numerous advantages. The key reasons include:

**1. Efficient Resource Utilization**

- Physical hardware resources such as CPU, memory, and storage are shared among multiple virtual machines.

- This maximizes the use of available resources and reduces hardware waste.

**2. Isolation and Security**

- Each virtual machine is isolated from others.
- If one VM crashes or is infected with malware, it does not affect the host system or other VMs.
- This provides a secure environment for testing and experimentation.

**3. Cost Reduction**

- Multiple virtual systems can run on a single physical machine, reducing the need for additional hardware.
- This lowers infrastructure and maintenance costs.

**4. Testing and Development Environment**

- Developers can test software on different operating systems without needing multiple physical devices.
- Virtual machines can be easily created, modified, or deleted.

**5. Flexibility and Portability**

- Virtual machines can be copied, moved, or backed up easily.
- They can run on different systems without compatibility issues.

**6. Support for Legacy Systems**

- Older operating systems that are no longer supported by modern hardware can still run inside virtual machines.
- This is useful for maintaining legacy applications.

**7. Disaster Recovery and Backup**

- Virtual machines can be saved as files and restored quickly.
- Snapshots allow recovery to a previous state in case of failure.

### 11.3 How Virtualization Works

Virtualization works through a combination of software and hardware mechanisms that simulate physical computing resources. The key components and processes are explained below:

**1. Hypervisor (Virtual Machine Monitor)**

The hypervisor is the core component of virtualization. It is responsible for creating and managing virtual machines.

There are two main types of hypervisors:

- **Type 1 (Bare-Metal Hypervisor):**
    - Runs directly on the physical hardware
    - Examples: VMware ESXi, Microsoft Hyper-V
- **Type 2 (Hosted Hypervisor):**
    - Runs on top of a host operating system
    - Examples: VMware Workstation, Oracle VirtualBox

In this project, VMware Workstation is used, which is a **Type 2 hypervisor**.

**2. Virtual Hardware Creation**

- The hypervisor creates virtual components such as CPU, RAM, hard disk, and network adapter for each virtual machine.
- These virtual components behave like real hardware devices.

**3. Guest Operating System Installation**

- The guest OS (NetRunner) is installed on the virtual hardware.
- It operates as if it is running on a physical computer.

**4. Resource Allocation and Management**

- The hypervisor manages how physical resources are distributed among virtual machines.
- It ensures fair and efficient allocation of CPU time, memory, and storage.

**5. Virtual Networking**

- Virtual machines can connect to networks using virtual network adapters.
- VMware provides different networking modes such as:
    - NAT (Network Address Translation)
    - Bridged Networking

- Host-only Networking

**6. Snapshots and Cloning**

- **Snapshots:** Save the current state of a VM and allow rollback.
- **Cloning:** Create copies of virtual machines quickly.

These features improve system recovery and experimentation.

### 11.4 Advantages of Virtualization in Modern Systems

- Improved hardware utilization
- Increased flexibility and scalability
- Enhanced system security
- Reduced operational costs
- Simplified system management

### 11.5 Limitations of Virtualization

- Performance overhead compared to physical systems
- Dependence on host system resources
- Limited access to certain hardware features
- Requires proper configuration and management

### 11.6 Summary

Virtualization is a powerful technology that enables multiple operating systems to run efficiently on a single physical machine. It improves resource utilization, enhances flexibility, and provides a secure environment for testing and development.

In this project, virtualization played a key role in successfully installing and running NetRunner on VMware Workstation. It demonstrated how modern computing systems use virtualization to simplify system management and improve overall efficiency.

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